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Rapid estimation of lithium-ion battery capacity and resistances from short duration current pulses

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HIGHLIGHTS

- SOH estimation is examined using voltage-capacity data during short-duration current pulses.
- Discharge capacity and pulse resistances at three SOCs are used as SOH indicators.
- A lightweight neural network produces rapid estimates of these seven SOH indicators.
- Method validation uses a publicly available NMC dataset and a new LFP dataset.
- Hardware-software integration is demonstrated by embedding the network onto a microcontroller.

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ABSTRACT

Rapid onboard diagnosis of battery state of health enables the use of real-time control strategies that can improve product safety and maximize battery lifetime. However, onboard prediction of battery state-of-health is challenging due to the limitations imposed on diagnostic tests so as not to negatively affect the user experience and impede normal operation. To this end, we demonstrate a lightweight machine learning model capable of predicting a lithium-ion battery's discharge capacity and internal resistance at various states of charge using only the raw voltage-capacity time-series data recorded during short-duration (100 s) current pulses. Tested on two battery aging datasets, one publicly available and the other newly collected for this work, we find that the best models can accurately predict cell discharge capacity with an average mean-absolute-percent-error of 1.66%. Additionally, we quantize and embed the machine learning model onto a microcontroller and show comparable accuracy to the computer-based model, further demonstrating the practicality of on-board rapid capacity and resistance estimation.

1. Introduction

The ever-increasing necessity for clean and renewable energy has placed significant pressure on the development and implementation of lithium-ion (Li-ion) batteries. From large scale applications like grid storage and electric vehicles to small consumer goods, Li-ion batteries are being used to fill the growing energy demand. This exigency for better energy storage brings with it the need for accurate and efficient means of battery health evaluation. The ability to estimate the health of Li-ion cells in operation is crucial to ensuring the longevity and safety of the systems they power. It also enables optimization of Li-ion battery recycling and repurposing for second-life applications [1–3].

The state of health (SOH) of a battery is often described by its remaining discharge capacity and internal resistance, both of which can be directly measured under controlled conditions [4–6]. Executing these measurements, however, is not always feasible for cells operating in the field as running a complete discharge cycle takes many hours and the cell resistance needs to be characterized at several states of charge (SOC). Furthermore, these quantities change with battery aging, and rerunning such tests to periodically characterize battery health requires significant downtime. Consequently, a strong necessity exists for battery SOH indicators like capacity and resistance to be estimated online using data from incomplete cycles or short test diagnostics to

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avoid interrupting regular usage and negatively impacting the user experience.

Recent studies have proposed a wide variety of SOH estimation methods, which can be generalized into two major categories: model-based methods and data-driven approaches. The former, being the more traditional approach, utilizes empirical models like an equivalent circuit model (ECM) [7–9], or physics-based approaches like electrochemical models [10–12]. The initially developed model is then updated in real-time using a state observer algorithm like the Kalman filter, one of its variants (e.g., the extended or unscented Kalman filter), or a particle filter. Despite the recent success, constructing these high-fidelity battery models is difficult and often requires conducting extensive materials characterization and cell-level testing over a wide range of conditions.

Data-driven models, on the other hand, can leverage large battery aging datasets to learn the complex relationships between extracted features and SOH metrics without the requirement of understanding the underlying physics. However, the success of these models is highly dependent on the quality of the data being used, making feature extraction a critical step in model development. Many researchers utilize voltage measurements during the constant-current (CC) charging stage for SOH estimation [13–18], while others have investigated incremental capacity [19,20], differential voltage [21,22], and even temperature [23] as model inputs, all with good results. However, these methods typically use data derived from CC charging events extracted from specific SOC ranges, which may not always be fully covered during battery operation. Further, the duration of continuous data required is generally greater than 10–15 minutes, which significantly limits the flexibility of cell operation. Researchers in [24] attempted to build a capacity estimation model that could still operate with missing data with relative success, yet their approach still required 5–10 minutes of CC charging data.

Despite data-driven approaches demonstrating satisfactory accuracy in SOH estimation, the data collection and practical implementation of feature extraction are often overlooked. In real-world applications, access to data at specific SOCs or of long duration is unfeasible, driving the need for rapid (<5 minute) SOH estimation methods. To achieve this, we propose a lightweight neural network (NN) to estimate cell capacity and internal resistance using the voltage response to a short-duration (100 seconds) current pulse. This study aims to reproduce the success of high-accuracy data-driven approaches while addressing their primary limitations of data acquisition and model complexity. The major contributions of this paper are summarized as follows:

- First, we propose a lightweight NN with a generalizable feature selection for SOH estimation. The model takes advantage of the underlying correlation between capacity and internal resistance to estimate cell SOH from a 100-second-long voltage response to a CC charge pulse.
- Second, we examine the model's performance on unseen-to-the-model cycling conditions, at different SOCs, and for charge and discharge pulse types. The model is validated on two datasets, one a publicly available NMC/Gr dataset, and the second a newly created dataset with 64 LFP/Gr 18650-cells.
- Third, we demonstrate the feasibility of real-world implementation on a stand alone system. The model is quantized and embedded in a custom developed, microcontroller-based charging system, providing real-time pulse application, data acquisition, and edge computation. Error from quantization is minimized and estimation accuracy is shown to be comparable to that of the un-quantized model.
- Fourth, we perform a feature importance analysis on the voltage response signal, and provide physical insight to the regions of the pulse most correlated with cell health metrics.

The rest of the report is organized as follows. In Section 2, background information is provided on current rapid SOH estimation methods. In Section 3, we describe the two datasets used in this work. In Section 4, the model architecture, input features, and evaluation metrics are defined. In Section 5, results on the model performance, feature comparisons, hardware implementation, and feature importance analysis are discussed. Finally, in Section 6, we summarize the major contributions and limitations of this work.

2. Background on SOH estimation

Health estimation of Li-ion batteries is a heavily studied field with recent work focusing on empirical health models and data-driven approaches. The vast majority of literature, however, relies on models developed using data that is restricted by SOC, requires a long diagnostic period, or requires specialized and expensive equipment to collect. To bridge the gap between online health estimation developed on highly controlled lab data and the random usage experienced in deployed batteries, rapid diagnostic models are being explored. Several promising approaches include using partial charging (PC) segments, the pulse test technique (PTT), and impedance based measurements [25].

2.1. Rapid SOH estimation using impedance measurements

Direct measurement of a cell's internal resistance over the full charge window requires a time-intensive diagnostic test, typically galvanostatic intermittent titration (GITT) or hybrid pulse power characterization (HPPC) [26]. A newer alternative, electrical impedance spectroscopy (EIS), provides detailed electrochemical information on the cell with a faster diagnostic time. EIS utilizes the application of a small amplitude alternating-current signal over a wide range of excitation frequencies to measure the cell's impedance and diffusion characteristics which reflect the state of the electrode active material and thus cell health [27]. Compared to other electrical-parameter-based SOH estimation approaches, EIS provides richer data suitable for both empirical and data-driven modeling [8,28–30]. Zhang et al. demonstrated using EIS for SOH estimation with a Gaussian process regression (GPR) [31]. Despite accurate capacity estimation, they showed that temperature and SOC can significantly affect the results, and the EIS test can still take several hours to perform. A common approach used in reducing the diagnostic time is to use the measured impedance values at only a few key frequencies, as opposed to using the entire spectra as model input [32,33]. The largest challenge with this method of SOH estimation is deconvoluting the relationship between EIS, cell internal resistance, and cell capacity since they can vary non-monotonically with aging [34,35]. While EIS spectra can be quickly collected, doing so requires special equipment for generating alternating currents in the cell. In turn, there remains a significant need to research new methods of rapid SOH diagnostics without the need for special equipment or long-duration test cycles.

Alternative approaches avoid the need for complex equipment by reverting to using direct-current (DC) excitation, as established with HPPC and GITT. However, instead of using a full charge characterization, DC pulses are applied at a few SOCs. This approach, commonly referred to as the pulse test technique (PTT), exploits the correlation between the measurable resistance during an applied DC pulse to a cell's capacity. Several studies have employed this approach to extract ECM model parameters [36,37] and for health estimation [1,38].

2.2. Rapid SOH estimation using voltage measurements

Direct current pulses also produce a transient response in the cell voltage that contains rich information about cell ohmic and ionic resistances due to mass transport and diffusion kinetics. Further, these cell resistances change with aging and have been shown to correlate well

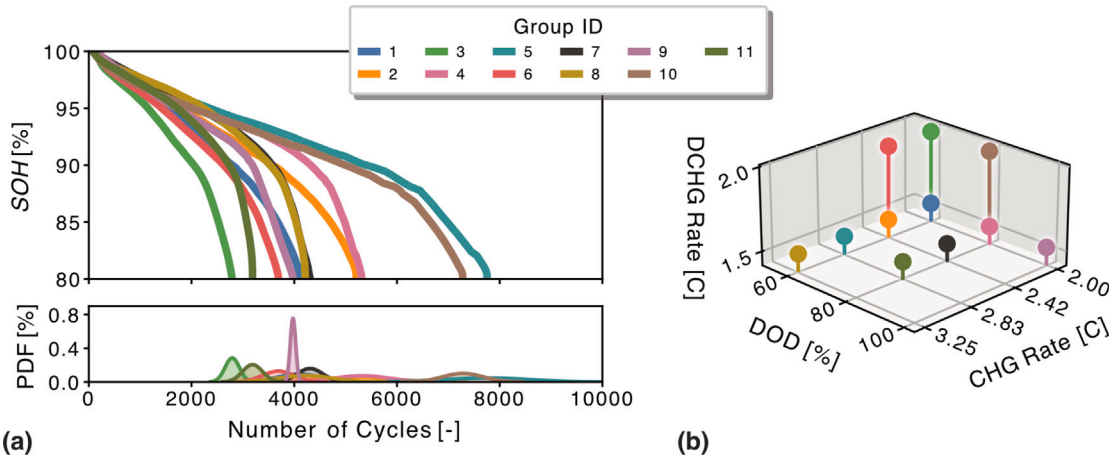


Fig. 1. Mean aging trajectories with end-of-life distribution (a) and cycling conditions (b) for each of the eleven groups within the LFP dataset.

with cell SOH [33,39,40], making them good candidates for diagnostic model inputs. A few researchers have used the knee points of the voltage response during DC pulses as a model input and have shown a strong correlation with discharge capacity. Ran et al. reported an accuracy of 95% when using a Gaussian process regression and showed that the high accuracy can be maintained even while estimating for batteries that are below 50% SOH [41]. Zhou et al. used pulse voltage response with the K-means algorithm to realize fast clustering of retired batteries based on their SOH [42]. Similar observations noted by other researchers using pulse currents show that the voltage profile collected during or immediately after pulse currents can be an effective indicator of battery health [43–45]. While the individual pulses are short, the requirement to take multiple pulses at different SOCs significantly increases the time of estimation. Another major limitation of these works is that the testing pulse is applied at a specific SOC interval or the end of the charge profile, thus making it infeasible for rapid estimation. Consumer batteries may not be subjected to a complete charge or discharge step thus leading to certain SOC ranges being inaccessible. Pre-processing of the voltage profile, which is required to generate the input features for the model, presents an additional level of complexity. First derivatives of the voltage profile which are commonly used to extract the significant features may result in the amplification of the noise present in the sensor readings. Lastly, the models are usually tested on a small number of batteries with limited variations in cycling conditions.

Partial charge models lend themselves towards rapid SOH estimation by utilizing features extracted from only a small segment of a charge cycle. Researchers have explored using portions of the CC and constant-voltage (CV) charging stages [46,47], however, the segment lengths are still on the order of 10 to 15 min. Tian et al. demonstrated capacity estimation using voltage segments between a lower and upper bound of 3.9 and 4.1 V [48]. The use of a fixed voltage window, however, introduces the condition that the cell be charged to a certain starting point before the diagnostic test can be run, removing the convenience of using a partial charge segment. Ruan et al. proposed using only 500 s during the CC–CV transition combined with a CNN to perform SOH estimation [49]. Compared to previous work, it greatly reduced the required sample size needed; however, ensuring the transient point between CC and CV stages in captured in the diagnostic test poses large challenges with practical implementation.

In this work, we contrast the partial charging method with a rapid SOH estimation model using only the raw voltage signal during a short duration (100 s) DC pulse. The model's effectiveness is shown at several SOCs to alleviate the need for significant cell conditioning prior to performing the diagnostic test and is tested on a diverse set of cycling conditions.

3. Datasets

Two datasets are used in this study. The first, newly created for this work, consists of 64 lithium-iron-phosphate/graphite (LFP/Gr) cells tested by Iowa State University and Iowa Lakes Community College, USA. The second dataset consists of 48 nickel-manganese-cobalt/graphite (NMC/Gr) cells tested by RWTH Aachen and Julich Aachen Research Alliance in Germany [39]. These two datasets (subsequently referred to as the "LFP" and "NMC" datasets, respectively) were chosen for their large number of cells and periodic application of pulses at different SOCs. The aging tests in both datasets consisted of constant-current cycling with a reference performance test (RPT) carried out at regular intervals. The RPT contains a sequence of pulses in charging and discharging steps aimed at measuring the internal resistance of the cells at different states of charge. It also measures the discharge capacity under a constant current protocol for all of the cells at a uniform C rate.

The cells used for the LFP dataset were LithiumWerks 18650 power cells with a nominal capacity of 1.2 Ah. Eleven groups were formed (6 cells in groups 1 through 10, and 4 cells in group 11), each with different cycling conditions. The eleven conditions consist of three depth-of-discharge (DOD) levels, two levels of discharge C rate, and four levels of charge C rate. Group aging trajectories and corresponding cycling conditions are shown in Fig. 1. The RPT was carried out once every 3.5 days (~100 cycles depending on the cycling conditions) and consisted of 0.2 C and 1 C current pulses at three SOCs: 20%, 50%, and 90%.

A large variation in aging trajectories is seen in the LFP dataset, with the number of cycles until the end of life (EOL) ranging from 2400 to 9000 cycles. For this work, the EOL is defined as the point at which that cell's discharge capacity drops below 80% of its nominal value. Group three cells are shown to age the fastest (mean EOL of 2800 cycles), with group five aging the slowest (mean EOL of 7900 cycles).

The NMC dataset was prepared using Sanyo/Panasonic UR18650E cells marked as C-grade by the manufacturer. These cells have a lower nominal capacity of 1.85 Ah when compared to the A-grade cells with a 2.05 Ah nominal capacity. All cells were subjected to the same cycling condition of 2C constant-current charge and discharge between approximately 20% and 80% SOC. An RPT was performed every 160 cycles, measuring the discharge capacity and pulse resistances at 30%, 50%, and 70% SOC. All cells were cycled below 50% SOH as seen in Fig. 6 of [39]. The deviation seen in the capacity trajectories, despite identical cycling conditions, was attributed to the variation in manufacturing. An overview of the datasets and their corresponding aging and RPT tests is given in Table 1.

Table 1
Cell descriptions and aging conditions for the LFP and NMC datasets. All cells are in the cylindrical, 18650 format. Here, the aging range for either dataset is defined according to SOH measured by cell capacity.

Chemistry	Capacity (Ah)	Cells tested	Cycling groups	Aging range (SOH)
LFP/Gr	1.2	64	11	1.0 to 0.8
NMC/Gr	1.85	48	1	1.0 to 0.6

4. Methodology

4.1. Model architecture and input/output features

Due to the inherently non-linear relationship between a cell's voltage profile and its SOH, an artificial neural network (NN) was chosen to map the relationship. The simplicity of a feed-forward neural network further reduces integration complexity when making the jump between research and commercial implementation.

This work explores two voltage-based model inputs for each of the two aforementioned datasets: time-series voltage segments during CC charging and DC internal resistance pulses. The prior is a well-established method in correlating the charge voltage to the SOH, and the latter is a rising focus in battery health prognostics. For both inputs, the NN was implemented using the *Keras Sequential* class. The model structure and hyper-parameters were selected via a series of grid-search optimizations. The optimization results and a discussion on the model training history are provided in Sections 1 and 2 of the Supplementary Material, respectively. The final model parameters are provided in Table 2.

To generate a set of voltage segments for model training and testing, the time-series voltage during CC charge (between 30% and 90% SOC) is sub-sampled in ten-minute segments with a maximum sample overlap of fifty percent. The voltage segments (sampled at 1 Hz) are mapped to an output consisting of seven parameters: the cell's remaining discharge capacity (Q_{dchg}), and the cell's internal resistances (R^{I}_{SOC}) for a charge/discharge rate of $I \in \{-1C, 1C\}$ at a SOC of $SOC \in \{20\%, 50\%, 90\%\}$ for the LFP dataset and of $SOC \in \{30\%, 50\%, 70\%\}$ for the NMC dataset. The modeling dataset was divided with a 2-to-1 train-test split, stratified by the cell's cycling conditions. That is, of the eleven cycling groups in the LFP dataset, seven were randomly selected for training, and the remaining four were used for testing. This stratification guarantees that the data used for testing was collected under cycling conditions unseen to the model, better reflecting in-field applications in which cell usage histories are often unknown. For the NMC dataset, samples were split by cell as all cells underwent the same cycling conditions. Note that we define a capacity-based SOH measure, denoted as SOH, and use this measure in all figures to allow for comparison between the LFP and NMC datasets. Mathematically, SOH is defined as

$$SOH = \frac{Q_{\text{dchg}}}{Q_{\text{nominal}}} \tag{1}$$

where Q_{dchg} is the cell's current discharge capacity and Q_{nominal} is its nominal discharge capacity.

The second model input consists of the time-series voltage (sampled at 1 Hz) during the applied DC pulse. In the case of the LFP dataset, the exact input is the concatenation of the cell voltage during a thirty-second C/5 charge, a 10-second pulse at 1C, and a subsequent one-minute rest. With each RPT containing pulses at each previously indicated SOC, there are a total of 16,728 individual pulse samples in the LFP dataset from which training and test sets are derived. The pulses in the NMC dataset were conducted without a preceding C/5 charge, thus the pulse voltage consists of a 1C charge for 20 s followed by a 30-second rest. The NMC dataset contains 3888 usable pulse samples. To remove slight variation in the starting voltage of a given SOC, the raw pulse voltage (V) is converted to a relative voltage (V_{rel}) with

$$V_{\text{rel}} = V - V_0 \tag{2}$$

where V_0 is the initial voltage during the C/5 charge step. This linearly shifts the pulse voltage while preserving information pertaining to cell degradation via changes in the slope. Note that for a given model, the train and test pulse voltages are drawn from a fixed SOC range. A model trained using pulses conducted at 20% SOC is tested only on pulses at the same SOC level. Further discussion is provided in Section 5.2.

4.2. Model evaluation metrics

The accuracy of the machine learning models developed in this work is evaluated with two error metrics: mean absolute percent error (MAPE) and root mean squared error (RMSE). The MAPE is reported as a percentage given by

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100 \tag{3}$$

where n is the sample size, y_i is the true output, and $\hat{y}_i$ is the predicted output obtained from the model. The RMSE retains the same units as the output feature (Ah for discharge capacity and mΩ for resistance), given by

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{4}$$

Unless specified, the reported RMSE and MAPE values are averaged over a 3-fold cross-validation.

4.3. Model quantization and optimization

The complexity of machine learning integration onto edge devices is heavily dependent on the model structure and size. Here, we embed the pre-trained neural network onto a custom-built programmable battery charger, controlled entirely by a single *Teensy 4.1* microcontroller. The *Tensorflow Lite* library is utilized to simplify the initial *Keras Sequential* model through a full eight-bit quantization process. A custom Python script then converts the quantized model to a byte array for inclusion in the microcontroller's firmware. The quantization process reduces the precision of the model's parameters, mapping the 32-bit floating point values to 8-bit integers.

This process, however, introduces additional prediction error that compounds with each floating-point operation performed during the model inference [50]. Model complexity is subsequently measured by the number of floating-point operations (FLOPs). To minimize the impact of quantization, achieving hardware prediction results that replicate the full model's accuracy, an optimization process is performed to reduce the size of the neural network. A simple sum-of-squares approach is used for the two optimization metrics (MAPE and FLOPs), given by

$$f_i = \sqrt{(\alpha * \tilde{x}_i)^2 + ((1-\alpha) * \tilde{y}_i)^2} \tag{5}$$

where the ith index of the cost function (f) is dependent on the normalized values of MAPE and FLOPs ($\tilde{x}$ and $\tilde{y}$, respectively) weighted by some scalar (α). The normalized values of MAPE and FLOPs are defined as

$$\tilde{x}_i = \frac{\text{MAPE}_i - \text{MAPE}_{\text{min}}}{\text{MAPE}_{\text{max}} - \text{MAPE}_{\text{min}}} \tag{6}$$

and

$$\tilde{y}_i = \frac{\text{FLOPs}_i - \text{FLOPs}_{\text{min}}}{\text{FLOPs}_{\text{max}} - \text{FLOPs}_{\text{min}}} \tag{7}$$

Note that subscripts min and max denote the minimum and maximum values of the given metric. In assigning a weight, α, a gridsearch

Table 2
Neural network parameters for the pulse voltage and segment of CC charge inputs.

Input feature	Input size	Hidden layers	Hidden layer size	Activation	Optimizer
Pulse	100	5	32	tanh	Adam
CC Segment	600	5	100	tanh	SGD

can be performed on various model structures to determine an optimal complexity. For this work, a weight of 0.8 was selected, preferentially biasing the optimized model towards higher accuracy. This decision is supported by the fact that a feed-forward, sequential neural network is not computationally complex relative to other machine learning models and can more easily be integrated on edge devices.

5. Results and discussion

In this section, we present comparisons of the pulse and CC segment input features, the type of pulse applied, and the SOC at which the pulse is conducted. A model optimization is performed with error metrics contrasted against several other common machine learning models. Application feasibility to battery management systems is shown via model quantization and implementation on a custom-developed hardware system.

5.1. Model dependence on input feature

Two main feature inputs are evaluated: the voltage response during a short-duration CC pulse and a voltage segment during a full CC charge. The CC voltage segments consist of ten-minute-long segments extracted from a 1C CC charge between 30% and 90% SOC, sampled at 1 Hz. This 600-second voltage segment is then used as an input to a neural network, outputting the estimated discharge capacity and six internal resistances at various states of charge. An overview of the CC segment sampling and the feature correlation with SOH is shown in Fig. 2(a–b). A similar feed-forward network is used with the pulse method. The voltage response during a sequence of charge steps, as described in Section 4.1, is used as the input. The pulse voltage has a length of 100 s, and the model outputs the same seven outputs, as seen in Fig. 2(d–e).

The input features are compared for their ability to accurately estimate the seven outputs. The parity plots in Fig. 2(c and f) show the diagnostic capabilities of each input feature, with the pulse input having a higher prediction density along the perfect prediction line. Only predictions for a single output (Q_{dchg}) are shown, however, a similar trend is seen among all seven outputs (Fig. 3). The reported MAPE values are averaged over three-fold cross-validation with error bars showing two standard deviations. A dummy model, predicting the mean value of the training samples, is used as a control for comparison against the using the pulse voltage and voltage segment during a CC charge. Note that the pulse input depicted in Fig. 3 represents a 1C charge pulse. Discussion of using a discharge pulse is provided in Section 5.2.

The pulse input is shown to outperform the CC voltage segment for all seven outputs and across both datasets. For the LFP dataset, Q_{dchg} is predicted with an average MAPE and RMSE of 1.6594% and 0.0235 Ah, respectively, when using the pulse input. The reduction in error is more prominent for the estimation of cell resistance, where a segment of CC charge produces an MAPE of 1.5863% and pulse inputs produce an MAPE of 0.4903% for charge resistance at 20% SOC.

Furthermore, the higher estimation accuracy seen in the NMC dataset, as compared to the LFP dataset, can be explained by the difference in dataset heterogeneity. The NMC dataset has a single cycling group, reducing the variation between train and test samples, whereas the LFP dataset consists of eleven different cycling groups. Despite the variation in cell usage history between the train and test samples, the pulse input is conclusively shown to better estimate cell health in comparison to using a voltage segment during CC charging.

Table 3
Estimation error of Q_{dchg} using a pulse input at 20% SOC when using different machine learning models. The MAPE (in %) and RMSE (in Ah) are provided, along with each model's hyperparameters. Only results from the LFP dataset are shown.

Model	MAPE (%)	RMSE (Ah)	Model parameters
ElasticNet	2.4479	0.0352	$\alpha = 1e-4$, $L_1 = 0.9$
Shallow NN	2.1923	0.0305	1 hidden layer with 32 neurons
Deep NN	1.6594	0.0235	5 hidden layers with 32 neurons
XGBoost	2.1650	0.0306	$\eta = 0.05$, max depth = 8
SVR	2.2632	0.0346	RBF kernel, $C = 1.0$, $\epsilon = 0.1$
GPR	2.2634	0.0313	Matern kernel, $l = 1$

5.2. Model dependence on pulse type

The type of applied pulse (charge versus discharge) has a marginal effect on predictive accuracy, as shown in Fig. 4. For all three states of charge used in the LFP dataset, the 1C charge pulse and equivalent discharge pulse had nearly identical MAPE values when estimating the cell's discharge capacity. Interestingly with the NMC dataset, using a charge pulse had a lower prediction error than using a discharge pulse. For pulses conducted at 30% SOC, the average MAPE values for estimating Q_{dchg} are 0.5344% and 1.0402% when using a charge pulse and discharge pulse, respectively.

Additionally, a parametric study is conducted to investigate how the SOC affects the accuracy in health estimation. Per Fig. 4(a), the MAPE remains fairly constant across all three SOC levels. The same trend is seen in Fig. 4(b) with the NMC cells, albeit under slightly different low and high SOCs. It should be noted that the assumed SOC of pulses collected in the LFP dataset vary from the true SOC with the deviation increasing as the cell ages. This arises due to the voltage-cutoff method used to determine SOC during the data collection stage. Further explanation is provided in Section 5 of the Supplementary Material. With practical implementation in mind, the motivation behind this methodology is to reduce the diagnostic time required. It is shown that a pulse can be conducted at any SOC with minimal impact on the estimation accuracy. Note, however, that the presented model uses train and test data at a single SOC. The model's performance on pulses collected on SOCs not included in the training set are provided in Section 4 of the Supplementary Material. The development of a truly SOC-agnostic model (i.e., one that accepts a pulse at an arbitrary SOC to predict the parameters of cell health) is reserved for future work.

5.3. Model comparison and hardware optimization

Further evaluation of the pulse feature is conducted on several common machine learning models. Table 3 presents the estimation error using ElasticNet, a shallow neural network, a deep neural network, XGBoost, support vector regression (SVR), and Gaussian process regression (GPR), with their associated model parameters. The deep neural network outperforms all of the other models, with the shallow neural network and GPR models performing similarly. The ElasticNet performs the worst with an MAPE of 2.4479%, a roughly 50% increase from the deep neural network.

Despite the greater accuracy produced with deeper neural networks, a trade-off needs to be made between computational complexity and estimation error. Chemali, et al. show that the computational time of a feed-forward neural network increases exponentially with the number of neurons per layer, and increases linearly with each additional hidden layer [51]. In this vein, an optimization between prediction error and

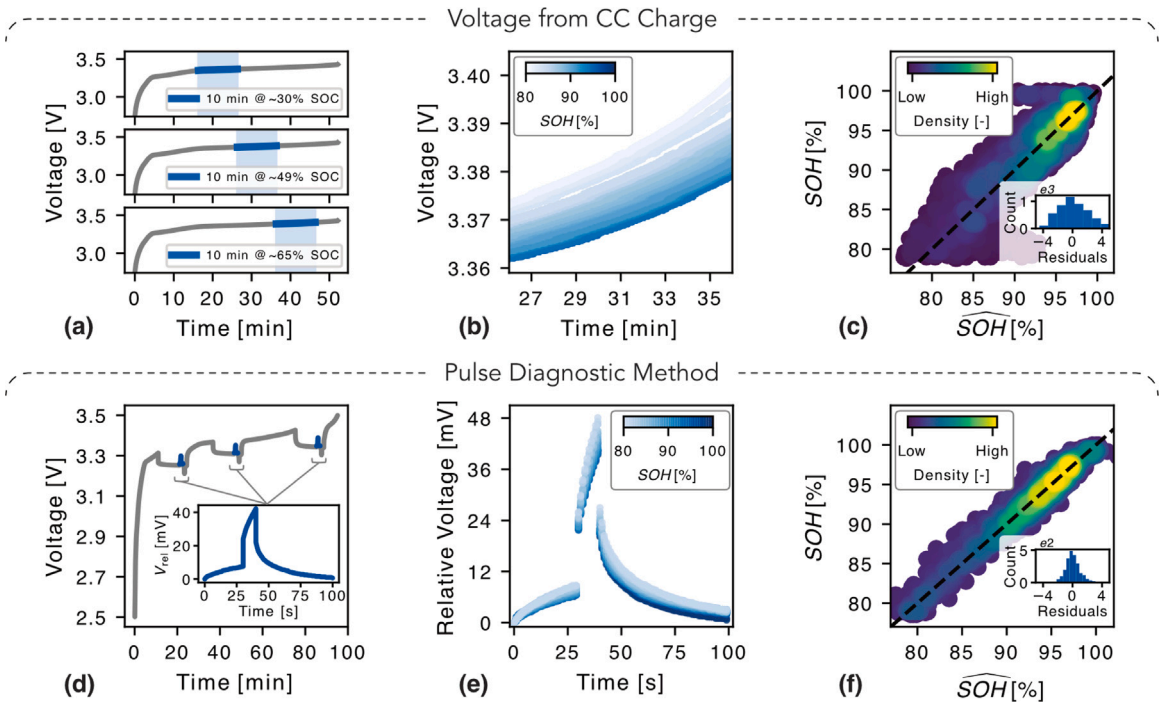


Fig. 2. An overview of the methodology for using a ten-minute segment of the voltage during CC charge (a–c) and the voltage during an applied DC pulse (d–f). The input feature sampling is shown in (a) and (d) with the feature correlation with cell SOH shown in (b) and (e). The resulting model prediction accuracy and error distribution are shown in (c) and (f). Data is shown only for the LFP dataset.

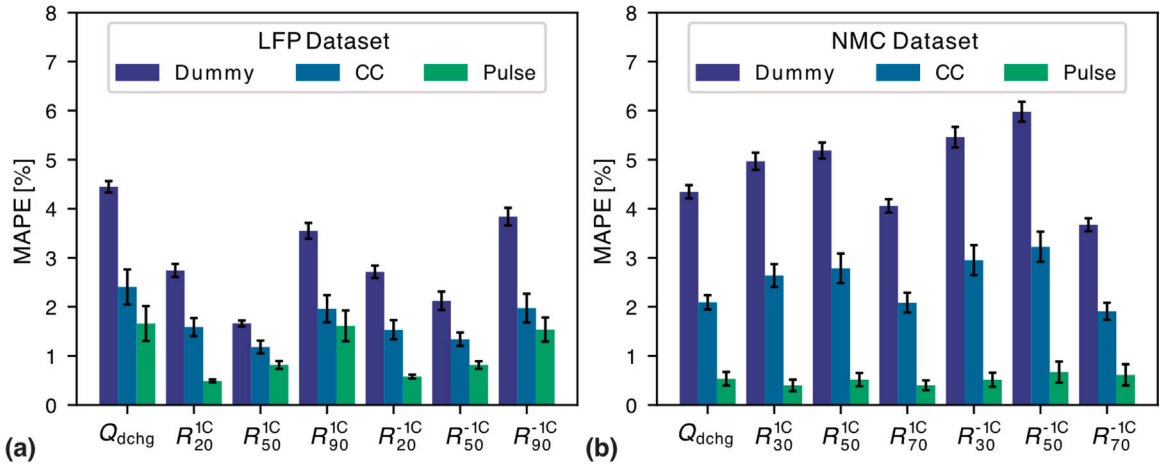


Fig. 3. Model prediction accuracy of the seven output parameters when using the voltage during a 1C CC charge and during a short duration pulse. Results are compared against a dummy model that predicts the mean value of each parameter for the (a) LFP dataset and (b) NMC dataset.

computational complexity was performed on the implemented deep neural network. By minimizing the cost function given in Eq. (5), an optimal number of hidden layers and neurons per layer can be determined. The results of the optimization with a bias weight, α, of 0.8 are shown in Fig. 5. A neural network of five hidden layers with 32 neurons per layer provided the best performance-to-complexity ratio. The reduced network size allows for easier implementation with edge computation.

5.4. Model quantization and hardware implementation

For embedded ML applications, the process of quantization is often required, reducing a model's activation weights from floating point values to integers. This reduction in data type size allows for faster computation, albeit at the cost of model accuracy. The loss due to quantization is illustrated in Fig. 6(a–b) where the pre-trained model's predictions are shown to have slightly lower error than those from the quantized model with an increase in MAPE of just under 4%. This process, however, enables the model to be easily implemented into hardware systems. Fig. 6(c) shows how the NN can be embedded into a microcontroller or edge device to perform battery health prognostics.

To test the feasibility of such an estimation method for real-world use, a custom hardware system was developed. The device, referred to here as an intelligent battery charger (IBC), leverages a unique hardware-software integration to provide rapid, in-field SOH estimation. A high-level overview of the IBC is shown in Fig. 7(a), which consists of a central microcontroller, a power supply, and a power module to regulate the output voltage and current. The digitally programmable output of the power module allows for a single microcontroller to perform all control logic and data processing, creating a low-cost and scalable system. A demonstration of the pulse application and data acquisition is shown in Fig. 7(b) via a serial printout of the

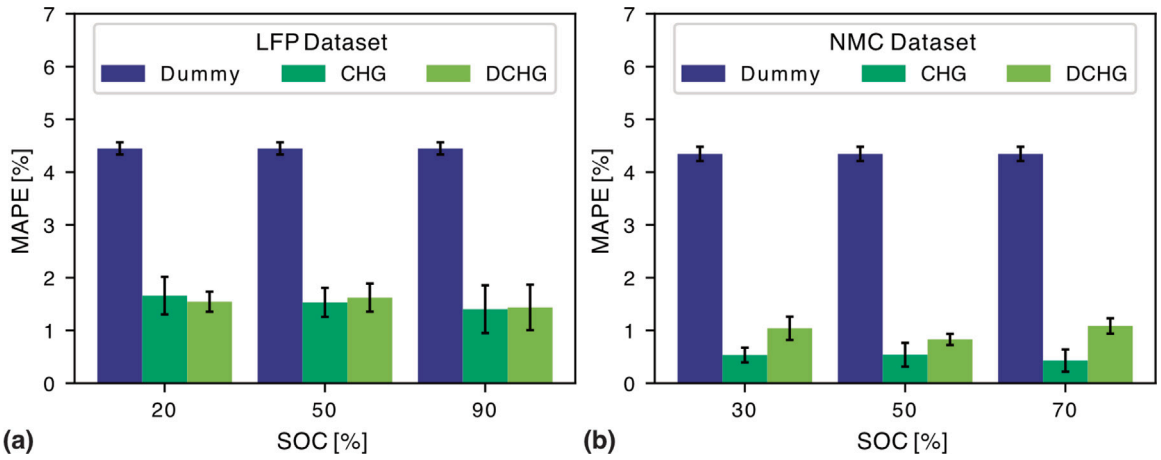


Fig. 4. Comparison between using a 1C charge or 1C discharge pulse input on estimation accuracy. Pulses from three states of charge are shown for the (a) LFP dataset and (b) NMC dataset.

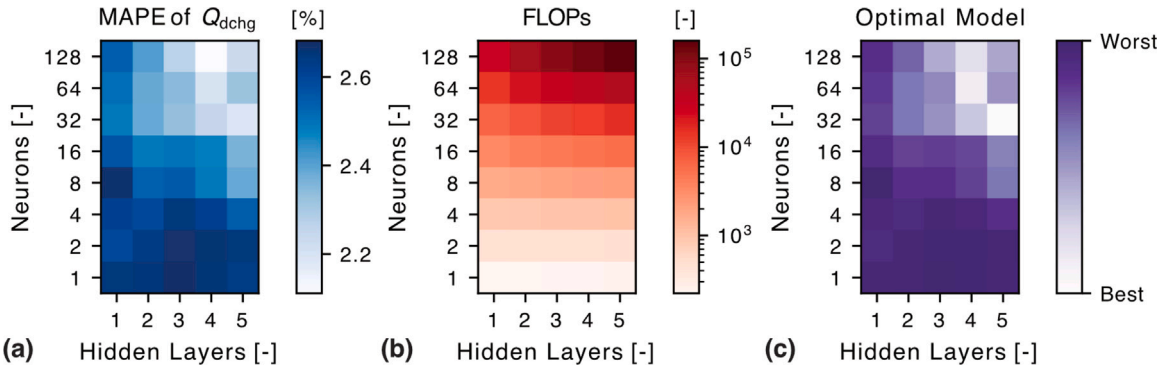


Fig. 5. An optimization of model performance and complexity. The model error (a), complexity (b), and optimization value (c) are shown as a function of the NN size.

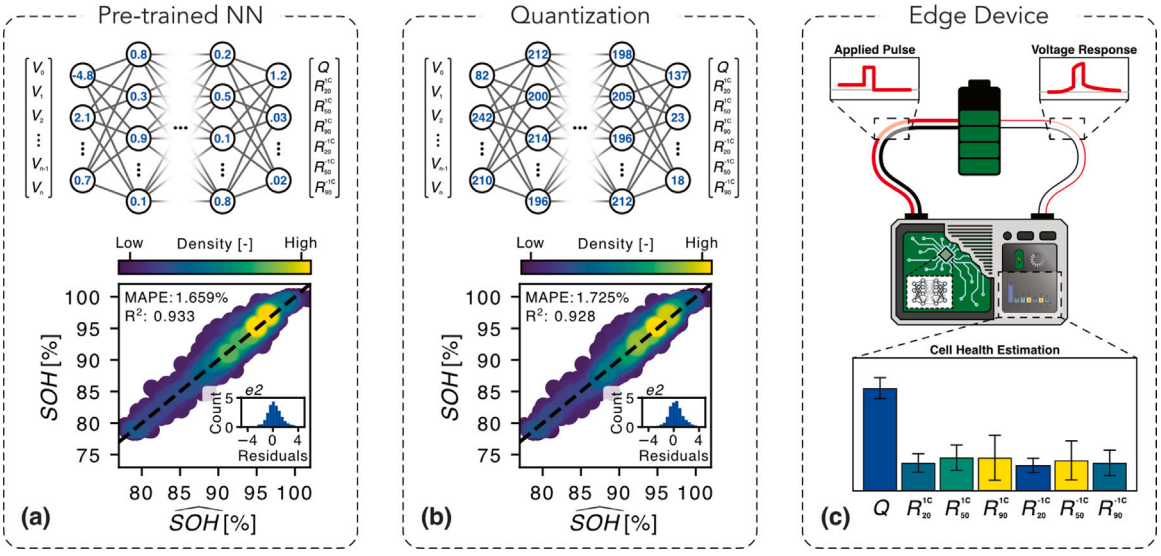


Fig. 6. A depiction of the quantization process including (a) the pre-trained model and prediction accuracy, (b) the quantized model and prediction accuracy, and (c) implementation on an edge device.

microcontroller's operation. The pulse is applied to a cell with the voltage and current being recorded at a user-set frequency of 1 Hz. The voltage response is then processed and interpolated per the model input requirements. The quantized model performs an inference on the cleaned voltage response, providing estimates of the cell's discharge capacity and six resistance parameters, as shown in step 6 of Fig. 7(b).

5.5. Feature importance

Thus far, we have only looked at model performance when using the full time-series voltage response. It is likely, however, that discrete points along the voltage signal are cross correlated and could be removed from the model input without losing significant predictive

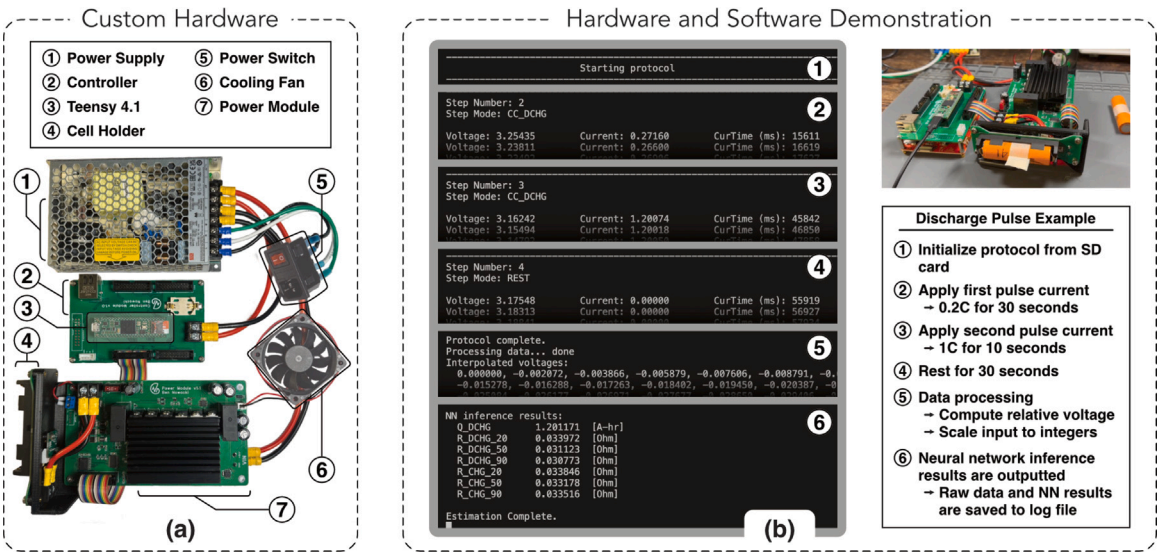


Fig. 7. Overview (a) and demonstration (b) of the custom-developed programmable battery charger.

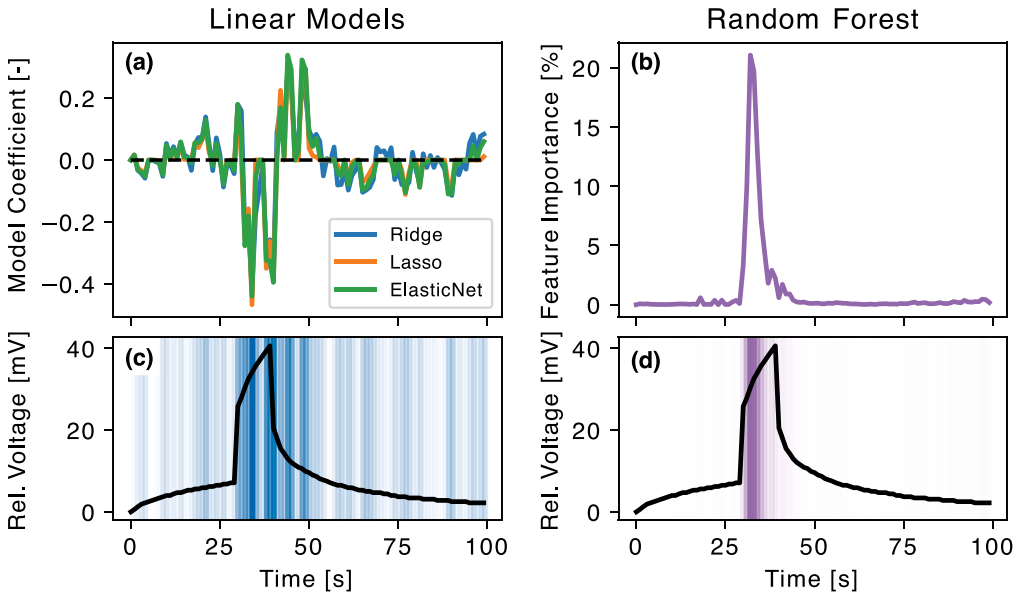


Fig. 8. Feature importance of each discrete voltage in the LFP dataset charge pulse. (a) Model coefficient values are compared across three linear models: ridge regression, lasso regression, and elastic-net regression. (b) Feature importance values of a random forest regression. (c) The magnitudes of the ridge regression coefficients are overlaid on the LFP charge pulse profile. (d) The random forest regression feature importance values are overlaid on the LFP charge pulse profile.

power. In this section, we explore the feature importance of discrete voltage points along the 100-second signal.

To evaluate feature importance, each voltage point is treated as a unique input feature. Fitting a linear model then allows us to extract the coefficients associated with each feature, providing insight into which portions of the voltage response are most correlated with cell health. A regularized regression provides the additional benefit of addressing the collinearity of several features with a penalty term [52,53]. The coefficients of three linear models (ridge regression, lasso regression, and elastic-net regression) and one non-linear regression (random forest) are evaluated. All model hyperparameters are first tuned using an iterative search of the parameter space using the same training and test split defined in Section 4.1. Note that these models are only trained using the charge pulse from the LFP dataset as input and only SOH as the target. The optimal model parameters and performance metrics are provided in Table S1 of the Supplementary Material.

As shown in Fig. 8a, all three linear models have similarly weighted coefficient vectors after training. It is important to note that the larger coefficients do not directly indicate a high correlation between SOH and the raw voltage at that discrete time, as the voltage signal is first standardized before being input into the model. Thus, the large coefficients indicate a relationship between SOH and the mean and variance of the voltage response at that time step. Interestingly, the results show clear regions of the pulse profile with large positive and negative coefficients (Fig. 8c). A potential explanation for this is the inherent relationship between the cell's voltage response to an applied pulse and its internal resistance. For a fixed applied current, an increased resistance will correspond to an increase in voltage, per Ohm's Law.

The most negative coefficients correspond to the peak voltage response during the 10-second 1C CC charge where an increase in internal resistance would be most noticeable. A higher internal resistance typically indicates a lower SOH, hence the negative coefficient. Similarly, the most positive coefficients align with the voltage relaxation period, where the exact relaxation rate depends on the internal charge transfer

resistances in addition to the cell's ohmic resistance [54,55]. A higher voltage immediately after the applied pulse would indicate a lower ohmic resistance, and consequently a higher SOH. A comparable pattern is observed in Fig. 8b and Fig. 8d with the feature importance values of the random forest model. Interestingly, the random forest model focuses primarily on the voltage response during the applied 1C CC pulse, giving almost no importance to the voltage relaxation period. From these coefficients and importance values, it can be shown that the model is primarily extracting the portions of the input signal most dependent on the cell's internal resistance. Short-duration current pulses highlight a cell's resistance in the voltage response, making it a powerful estimation method for rapid SOH diagnosis.

6. Conclusion

Rapid and accurate estimation of discharge capacity and resistances are crucial to the development of accurate battery management systems. This work demonstrates a technique using the voltage obtained during a short-duration current pulse as the input to predict the capacity of a battery. The validity of the technique is shown by applying it to two different datasets with different chemistries, LFP and NMC, and across various cycling conditions. The observed MAPE of capacity estimation using the pulse voltage was ~1.5% for the LFP dataset compared to ~2.5% when using the voltage during a CC charge. The accuracy disparity is more prominent in the NMC dataset with a MAPE of ~0.5% and ~2.0% for the pulse method and CC segment, respectively. The pulse voltage data is minimally processed before being fed into the model, drastically reducing the overhead required for feature extraction. By comparing the errors associated with different pulse types and SOCs, it was shown that the technique is agnostic to the point of application of the pulse. Furthermore, the model was quantized and incorporated into a microcontroller to test its real-life applicability. A custom testing setup capable of applying pulse currents and measuring voltage response was built. The quantization loss was minimized by optimizing the number of hidden layers and neurons with the estimation accuracy, demonstrating a simple system with practical applications.

CRediT authorship contribution statement

Benjamin Nowacki: Writing – review & editing, Writing – original draft, Visualization, Software, Hardware, Methodology, Investigation, Formal analysis. **Jayanth Ramamurthy:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis. **Adam Thelen:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Chad Tischer:** Investigation. **Cary L. Pint:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Chao Hu:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jpowsour.2024.235813.

Data availability

Code and processed data for generating all of the results and figures are available on GitHub. A link to the newly collected LFP/Gr dataset will be added to the above repository soon after publication.

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